

Lord of the Ring – The Arena Returns

Summary Report

IoT Applications Club (IoTA),
Jadavpur University, Kolkata, India

Abstract - Lord of the Ring: The Arena Returns is the third edition of the flagship Robo-Sumo competition organized by the IoT Applications Club (IoTA), Jadavpur University, presented at Srijan 2026 - the Annual Techno-Management Fest of Jadavpur University. Held across two days on April 10-11, 2026 at the Salt Lake Campus, the event drew participation from over 40 teams and approximately 165 registered participants from institutions across West Bengal and beyond. The competition featured two distinct categories - Wired and Wireless - with a combined prize pool of Rs.12,000. This report presents a comprehensive overview of the event, covering competition format, rules, participating teams, results, and institutional footprint.

I. INTRODUCTION

The Internet of Things (IoT) has become a defining force in modern engineering, enabling intelligent systems that bridge the physical and digital worlds through real-time sensing, communication, and data-driven decision making. Robotics, as an applied branch of this ecosystem, offers one of the most tangible and intellectually stimulating arenas for students to develop hardware design, embedded systems, and control engineering skills.

The IoT Applications Club (IoTA) at Jadavpur University is the official student-run technical society dedicated to exploring and advancing the IoT and robotics ecosystem. Founded in 2023, IoTA operates as a collective of visionary engineers united by a shared mission: to bridge the gap between theoretical knowledge and practical hardware innovation. The club is supported by the IEEE Circuits & Systems Society (IEEE CAS), Jadavpur University Chapter, and operates under the mentorship of senior faculty from the Department of Electronics & Telecommunication, Jadavpur University.

IoTA's activities span three core pillars: (1) Advanced R&D — developing low-latency sensor networks and embedded smart systems; (2) Knowledge Fusion — conducting deep-dive workshops on Embedded C, PCB design, and RTOS to make members industry-ready; and (3) Robotics & AI — merging mechanical precision with intelligent algorithms to build autonomous, perception-driven systems.

Since its inception, IoTA has organized a range of technical events including Connexion 2025 (an IoT ideation competition), Robotic Workshops, PCB Design Bootcamps, and the flagship Robo-Sumo competition — Lord of the Ring — now in its third edition. The club's vision is to become the epicenter of IoT research and product development at Jadavpur University and to foster a culture of radical technical excellence and professional growth among students.

II. ABOUT SRIJAN 2026 & LORD OF THE RING

Srijan is the Annual Techno-Management Fest of Jadavpur University, organized by the Faculty of Engineering and Technology Students' Union (F.E.T.S.U.). It is one of the premier collegiate technical festivals in Eastern India, bringing together students, innovators, and industry professionals from across the country for competitions, exhibitions, and workshops.

Lord of the Ring — 'The Arena Returns' — is IoTA's flagship event at Srijan 2026, and represents the third consecutive edition of this Robo-Sumo competition. The event pits custom-built robots against each other in an elimination-style arena combat, challenging participants to demonstrate not only engineering design skill but also real-time strategic decision-making and control precision.

The 2026 edition was held on April 10–11, 2026, at Jadavpur University Salt Lake Campus, Kolkata. Registrations were accepted through both the Srijan official portal (www.srijanju.in) and the Unstop platform, with the last date for registration being April 5, 2026. The event was open to teams of 3–5 members each, with a total prize pool of ₹12,000 distributed across two independent categories.

III. COMPETITION FORMAT & CATEGORIES

A. Event Timeline

Date	Category	Details
April 10, 2026	Wireless Category	Full Event — Salt Lake Campus, JU
April 11, 2026	Wired Category	Full Event — Salt Lake Campus, JU

B. Categories

- **Wireless Category:** Robots powered by onboard batteries, controlled wirelessly via remote. Initial rounds held on April 10 and finals on April 11, 2026. Distinct prizes awarded.
- **Wired Category:** Robots powered by a 220V AC wired supply, with extended wires strongly recommended. Full event conducted on April 11, 2026. Distinct prizes awarded.

IV. GAME FORMAT & RULES

A. Arena Specification

The competition arena is a circular platform divided into three concentric zones, as illustrated in Figure 1:

- **Red Zone (Combat Area):** The central circular combat zone with a radius of 2 ft. This is the primary battleground where robots engage.
- **Green Zone (Danger Zone):** A 1 ft wide ring surrounding the Red Zone, with a total outer radius of 3 ft. Entering this zone signifies strategic danger for the defending robot.
- **Elimination Zone:** The area entirely outside the Green Zone. Any robot displaced into this zone is considered eliminated from the round.

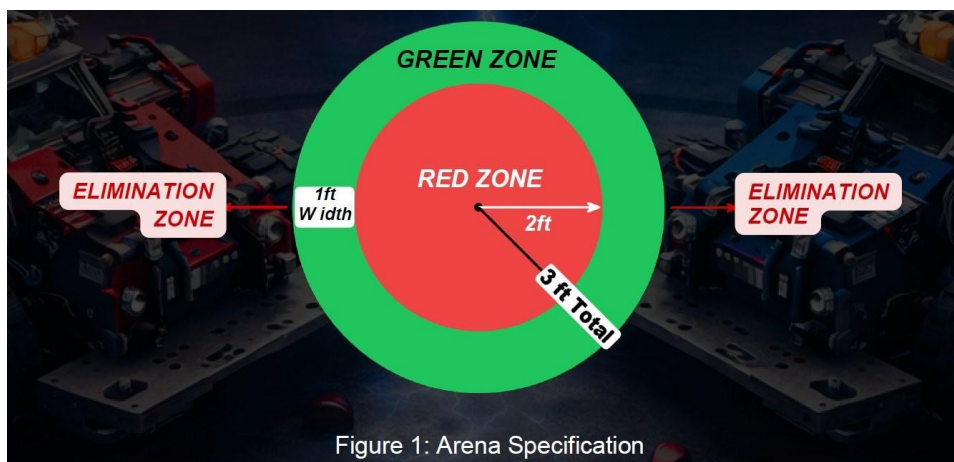


Figure 1: Arena Specification

Figure 1: Arena Specification

B. General Rules

- **Team Size:** Each team must consist of 3 to 5 members.
- **Categories:** Wired and Wireless are treated as completely separate competitions with independent prize pools and brackets.
- **Format:** Elimination rounds — a losing team exits the competition immediately after defeat.
- **Match Duration:** Each match consists of 3 rounds, each lasting 1 minute.
- **Victory Condition:** The first robot to win 2 out of 3 rounds wins the match. Winners are also determined based on strategic moves and positioning.
- **Judges:** All decisions regarding task completion, movement precision, and strategic execution rest with the designated IoTA club judges. In all disputes, the judges' decision is final.

C. Scoring System

- **+2 Points — Green Zone Push:** Awarded when a robot successfully pushes the opponent into the Green Zone (danger ring).
- **+5 Points — Elimination:** Awarded for complete displacement — pushing the opponent robot fully outside the arena into the Elimination Zone.
- **Tiebreaker:** If both robots exit the arena simultaneously, the robot that leaves the arena later is awarded points based on the sector of exit.
- **Penalty:** Intentionally entering the arena before the official start signal incurs negative points against the offending team.

D. Technical Specifications

Parameter	Specification
Maximum Dimensions	30 cm × 30 cm × 15 cm
Maximum Weight	2.5 kg
Minimum Ground Clearance	7 mm
Maximum Wheel Diameter	6 cm

Parameter	Specification
Allowed Motors	High Torque PM Brushed DC Motors, Standard Gearboxes
Banned Motors	BLDC, Stepper Motors, Industrial Servos
Banned Gearboxes	Planetary Gearboxes, Cycloidal Drives
Banned Mobility	Mecanum/Omni Wheels, Sticky Tires, Suction Systems

E. Mandatory Safety Rule — The Heart Rule

A 20A Automotive Blade Fuse with a holder **MUST** be installed at the robot's power line. This fuse acts as the high-current cap for the entire drive system. If the fuse is blown during a match, the robot is considered mechanically disabled and forfeits the round. Teams are strongly advised to maintain an on-site inventory of critical replacement components to minimize downtime between rounds.

V. PRIZE POOL

The total prize pool for Lord of the Ring 2026 was ₹12,000, distributed equally and independently across the two categories:

Position	Wired Category	Wireless Category
1st Prize	₹ 3,000	₹ 3,000
2nd Prize	₹ 2,000	₹ 2,000
3rd Prize	₹ 1,000	₹ 1,000
Category Total	₹ 6,000	₹ 6,000

Each category is rewarded independently, ensuring that both the Wired and Wireless participants compete for equal prize stakes.

VI. PARTICIPATING TEAMS & REGISTERED PARTICIPANTS

Registrations were accepted via two platforms: the official Srijan 2026 portal and the Unstop platform. After de-duplication of teams registered on both platforms, the total participation count was as follows:

Metric	Count
Total Finalized Teams	~43
Total Registered Participants	~165
Wired Category Teams	16
Wireless Category Teams	32
Teams in Both Categories	~14

A. Wired Category Teams (Participated on 11th May)

Sl No.	Team Name	Members
1	Ares 2.0	Shirsendu Kundu, Amit Mondal, Abhijit Mahata
2	Ares 3.0	Megh, Sourashis Das, Soumil Majumdar
3	Ares (Wireless)	Bishal Tarafder, Arin Sen, Somnath
4	Titanium Robotics 1.0	Saptangshu Roy, Sounak, Ishika
5	Titanium Robotics 2.0	Sudipta, Aniket, Barnali Roy
6	Titanium Robotics 3.0	Shuravi, Deep, Sujit
7	RoboMeow	Tejas, Hricha Samanta, Ishaan Chowdhury, Sourjya Ghosh, Swapnaneel Chattaraj
8	Rocky (Helios)	Spandan Das, Akash Maji, Prakriti Ghosh, Pubali Ray, Kaustav Karmakar
9	Black Pearl Robotix	Suraj Shaw, Shubham Panjiyara, Muskan Jaiswal
10	Cache Crushers	Soumik Das, Soham Mukherjee, Ranit Nandi
11	Platinum Crest	Aditya Kumar Chaurasia, Vivaan Chakraborty, Rayirth Roychoudhury, Jay Vardhan Bagaria
12	Excalibur	Rudraksha Podder, Swastik Chakraborty, Arihan Das, Gritav Sultania, Nuwan Jaiswal
13	The Apex Predators	Aghnya Kundu, Sourasish Mitra, Mayukh Choudhury, Avirup Majumder, Srijita Murmu
14	Broke Builders	Subhayan Ghosh, Soumalya Jar, Bubai, Arpan Das
15	Black Pearl Robotix	Suraj Shaw, Shubham Panjiyara, Muskan Jaiswal
16	Nexus	Subhrajyoti Ghosh, Dipayan Das, Debaditya Das

B. Wireless Category Teams (Participated on 10th May)

Sl No.	Team Name	Members
1	Fat Boy	Rwitaban Sengupta, Swastik Saha, Arka Prava Das, Mayank Jindal
2	HERACROSS_0.1	Nilesh Saha, Ankita Acharya, Binayak Bisoi
3	DHUMKETU	Sibsankar Mondal, Ayan Choudhury, Shuvadip Dey, Nirupam
4	Horsepower	Review Mondal, Rohit Saha, Suprateek Jash
5	Team X	Maaz Ahmad, Md Rakib, Azmaen, Shawaiz Jafry
6	Shinra Tensei	Shreyam Marodia, Apurba Mandal, Saptashwa, Sushmit Mitra, Shirsho Saha
7	MAVERICK	Anujoy Barman, Protiva Saha, Anwit Patra, Deepchand Banik
8	Titanium Robotics 1.0	Saptangshu Roy, Sounak, Ishika
9	GiganigaBytes	Sagnik Taraphder, Pratik Dolai, Snehadip Ghosh, Agniva Saha Roy, Yexomate
10	VoltEdge	Manish Das, Rishi Chakrabarti, Moulik Ghosh, Anurag Singha, Dipanjan Dey

Sl No.	Team Name	Members
11	Titanium Robotics 2.0	Sudipta, Aniket, Barnali Roy
12	Mechasaurus	Sitabja, Ashif Ansari, Arya Biswas, Tiya Maiti
13	YANTRIK	Priyabrata Sur, Anway Ghatak, Archis Pal, Puskar Malik, Srijan Dey
14	Asphalt Drifters	Samya Dolai, Subhadeep Barua, Avijit Mondal, Partha Khanra
15	RC Crushers	Somyajit Ram, Debarjyo Ghosh, Tanzeem Aziz Ansari, Bhaskar Das
16	Ares 3.0	Megh, Sourashis Das, Soumil Majumdar
17	RoboMeow	Tejas, Hricha Samanta, Ishaan Chowdhury, Sourjya Ghosh, Swapnaneel Chattaraj
18	Titanium Robotics 3.0	Shuravi, Deep, Sujit
19	Wava Gang	Aaditya Kumar, Soumadeep Mondal, Animesh Prasad, Mayank Singh, Rohit Naskar
20	SumoC	Pritish Dutta, Debneel Ghosh, Ekant Kumar, Sirshendu Dutta, Eshan Mondal
21	The Apex Predators	Aghnya Kundu, Sourasish Mitra, Mayukh Choudhury, Avirup Majumder, Srijita Murmu
22	RoboWarriors 1.0	Ankur Karmakar, Biswajit Karan, Rohit Kumar Roy, Soumyajit Boral
23	Cache Crushers	Soumik Das, Soham Mukherjee, Ranit Nandi
24	Diamond Blender	Gulam Subhani, Sunil Kumar Mehta, Ayush Kumar Ojha
25	the final chapter	Subinoy Haldar, Rajesh Roy, Aritra Biswas, Saikat Dey, Arman Iqbal
26	BlackBots	Soumyaditya Agarwalla, Freya Kanungo, Agneesh Pramanik, Shalini Das,
27	AXON	Ayan Chandra, Tanav Bhattar, Anik Acharya, Mohammed Arsalan
28	PHANTOM SQUAD	Krish Kumar Burnwal, Supriyo Mondal, Abhijeet Baraik, Ayan Das,
29	MechaRex	Sohan Bhadra, Swarnava Paul, Adrija Biswas
30	Nightcrawler	Rajarshee, Oishee Halder, Aritra Das Choudhury, Krrish Singh
31	Rust in Peace	Sayan Sill, Malyadeep Sasmal, Shubham Panjiyara
32	OTC	Mayank Jindal, Swapnendu Chatterjee, Aadil Farooq

VII. INSTITUTIONAL FOOTPRINT & PARTICIPANT DIVERSITY

Lord of the Ring 2026 attracted participation from institutions across West Bengal and multiple other Indian states, underscoring the national reach of the event. The table below presents the college-wise breakdown of registered participants:

#	Institution	Location	Participants
1	Jadavpur University (JU)	Kolkata, West Bengal	~86
2	B.P. Poddar Inst. of Mgmt. & Tech. (BPPIMT)	Kolkata, West Bengal	12
3	Adamas University	Kolkata, West Bengal	9
4	Heritage Institute of Technology (HITK)	Kolkata, West Bengal	5
5	Future Institute of Technology (FIT)	Kolkata, West Bengal	5
6	Academy of Technology (AOT)	Hooghly, West Bengal	5
7	Inst. of Engineering & Management (IEM)	Kolkata, West Bengal	4
8	Kalyani Government Engineering College (KGEC)	Nadia, West Bengal	4
9	Techno Main Salt Lake	Kolkata, West Bengal	4
10	University Institute of Technology, BU	Burdwan, West Bengal	4
11	Indian Maritime University, Kolkata	Kolkata, West Bengal	3
12	Govt. College of Engg. & Textile Technology	West Bengal	1
13	Welingkar Institute of Management	Mumbai, Maharashtra	3
14	Graphic Era University	Dehradun, Uttarakhand	2
15	IIT Patna	Patna, Bihar	1
16	IIT Madras	Chennai, Tamil Nadu	1
17	Schools & Others (Purwanchal, DPS, Salt Lake, etc.)	West Bengal	~9

Of the total ~165 participants, approximately 52% were from Jadavpur University, 38% from other West Bengal institutions, and 10% from institutions outside West Bengal — spanning Maharashtra, Bihar, Uttarakhand, Tamil Nadu, and Delhi. This geographic diversity reflects the growing national visibility of IoTA's flagship event.

The event also showcased the inclusivity amongst future engineers, as 2 teams from **Class VIII-X participated into the event & showcased excellent skills in building & fighting in Lord of The Ring. Team Platinum Crest & Team Excalibur, consisting of little brothers from Kolkata's top high school institutions took part in the wired segment.** This showcases the commitment of IoTA & IEEE CAS to involve community & future generations to practical learning.

VIII. RESULTS & WINNERS

A. Wireless Category

Position	Team Name	Members
1st	Black Bird Robotix	Ayan Kumar Mahapatra, Rajdeb Mukhopadhyay, Rhitam Biswas, Arman Ali, Abhronil Naskar, Ankan Bera
2nd	Ares	Bishal Tarafder, Arin Sen, Soumil Majumdar, Shirshendu Kundu
3rd	Titanium Robotics 1.0	Saptangshu Roy, Sounak, Suravi Shaw

B. Wired Category

Position	Team Name	Members
1st	Ares 2.0	Shirsendu Kundu, Amit Mondal, Abhijit Mahata
2nd	Blackbird 1.0	Ayan Kumar Mahapatra, Rajdeb Mukhopadhyay, Rhitam Biswas, Arman Ali, Abhronil Naskar, Ankan Bera
3rd	Nexus	Subhrajyoti Ghosh, Dipayan Kundu, Khokon Barman, Debadrito Chowdhury

IX. EXPENDITURE



X. SPONSORS & PARTNERS

Lord of the Ring 2026 was made possible through the generous support of the following sponsors and institutional partners:

Partnership Type	Sponsor / Organisation
Academic Partner	IEEE Circuits & Systems Society, Jadavpur University (IEEE CAS JU)
Platinum Partner	Wisdom Kids
Hardware Partner	Haque Electronics
Industry Partner	ElectroMAF India
Learning Partner	Easy2Learning

XI. ACKNOWLEDGEMENTS

We would like to express our sincere gratitude to the University administration of Jadavpur University for providing the venue and institutional support for organizing Lord of the Ring 2026 at the Salt Lake Campus.

We extend our heartfelt thanks to the faculty mentors of the IoT Applications Club whose guidance and continuous support have been the backbone of all IoTA initiatives:

- Prof. Sheli Sinha Chaudhuri — Professor, Dept. of Electronics & Telecommunication, Jadavpur University
- Prof. Palaniandavar Venkateshwaran — Professor, Dept. of Electronics & Telecommunication, Jadavpur University
- Dr. Chinmoy Ghorai — Assistant Professor, Dept. of Electronics & Telecommunication, Jadavpur University
- Prof. Chandrima Mondal — Professor, Dept. of Electronics & Telecommunication, Jadavpur University

We are also deeply grateful to the organizers of Srijan 2026 and F.E.T.S.U. for providing the platform to host this event at one of Eastern India's premier technical festivals.

Finally, we acknowledge every participant, volunteer, and IoTA club member whose dedication, hard work, and passion for robotics made Lord of the Ring 2026 an electrifying and memorable competition.

Gallery



Pic 1: Champion & Runnerup Teams of Lord of the Ring (wireless)



Pic 2: Arena Setup of Lord of The Ring (IEEE CAS logo at the flex & arena)



Pic 3 : Team Platinum Crest (High school team) with Team Ares after match. Withholding spirit of inspiration & innovation of IEEE CAS.



Pic 4 : Team Maverick & Team Wava Gang before the Group Stage Round.



Pic 5 : Round of 16 fighting ongoing in the Arena. (CAS Logo visible at top)



Pic 6 : Talks before prize distribution ceremony



Pic 7 : Team Ares receiving 1st Prize for Lord of The Ring (Wireless)



Pic 7 : Team Nexus receiving 3rd Prize for Lord of The Ring (Wired)



Pic 7 : Team Blackbird Robotics receiving 2nd Prize for Lord of The Ring (Wireless)



Pic 7 : Team Titanium Robotics receiving 3rd Prize for Lord of The Ring (Wireless)



Pic 8 : Lord of The Ring Poster

